SCENARIO-BASED QUESTIONS

Computer Vision / Image Processing

Complete answers for Questions 3, 4 and 5

SIMATS Engineering|Computer Vision(ITA-0521)

Assessment Tool:2

Includes basic concepts, detailed answers to parts (a)–(d), step-by-step procedures, worked numerical examples, comparison tables, illustrative diagrams, short exam-ready answers, conclusions and extra viva questions.

Overview of the three image-processing tasks



Note: The explanations and examples below are prepared for study and assessment. Actual results depend on the image and filter settings.

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Question 3: Salt-and-Pepper Noise Removal

Scenario: A scanned document has small black and white dots scattered over it. The text is hard to read.

1. The scenario and the questions

(a) What is the noise called and what does it look like?
(b) What problems does it cause?
(c) Which filter removes it?
(d) Why is it better than an averaging filter?
(e) Write your answer step by step.

2. Basic ideas you must know first

Digital image: A digital image is made of pixels. In a grayscale image, pixel values commonly range from 0 to 255. A value of 0 represents black, 255 represents white, and values between them represent shades of gray.

Noise: Noise is an unwanted change in pixel values. Salt-and-pepper noise affects selected pixels by changing them to extreme black or white values, while many neighbouring pixels remain correct.

Salt-and-pepper noise: It is also called impulse noise. White pixels are called salt and black pixels are called pepper.

Name	Pixel value	Appearance	Common place noticed
Salt	255	Small white dot	Inside dark letters or objects
Pepper	0	Small black dot	On white paper or bright areas

Why this matters: Since the corrupted pixels are extreme values, a method that uses the middle neighbourhood value can remove isolated dots without mixing them into nearby pixels.

3. Answer to (a): What is the noise called and what does it look like?

The noise is **salt-and-pepper noise**. It looks like random black and white specks sprinkled over the document. White specks may break dark letters, while black specks appear on the white paper. Possible causes include dust or dirt on the scanner, faulty scanner elements, and errors during image transmission or storage.

4. Answer to (b): What problems does it cause?

• **Reading difficulty:** dots cover parts of letters and make words less clear.

• **Broken or thick letters:** a white dot can create a hole in a dark letter, while a black dot can make a letter look thicker.

• **Incorrect punctuation or numbers:** isolated dots may be mistaken for full stops, commas or decimal points.

• **OCR errors:** Optical Character Recognition may miss characters or recognise extra symbols.

• **Poor image quality:** the scanned page looks dirty and becomes less suitable for printing or archiving.

5. Answer to (c): Which filter can remove it?

Use a **3 × 3 median filter**. The filter examines a pixel and its eight neighbours, sorts the nine values, and replaces the centre pixel with the middle (fifth) value. A 3 × 3 window is usually a good starting point for light or moderate noise.

5.1 Steps performed by the median filter

Step 1: Place the window. Put a 3 × 3 window over the pixel to be corrected.

Step 2: Collect values. Read the centre pixel and its eight neighbours.

Step 3: Sort values. Arrange the nine values from smallest to largest.

Step 4: Choose the median. Take the fifth value in the sorted list.

Step 5: Replace the centre. Use the median as the corrected centre-pixel value.

Step 6: Repeat. Slide the window across the image and repeat for all required pixels.

6. Answer to (d): Why is it better than an averaging filter?

An averaging filter adds the neighbourhood values and divides by the number of pixels. Extreme values such as 0 and 255 can strongly pull the average away from the true local brightness. The median filter selects the middle value instead, so isolated extremes usually do not affect the result. It removes dots while better preserving the sharp boundaries of printed letters.

Property	Averaging filter	Median filter
Calculation	Adds values and divides by 9	Sorts values and selects the fifth
Effect of isolated 0 or 255	Can pull the output far from the correct value	Usually ignored when surrounded by normal values
Salt-and-pepper noise	May turn dots into gray smudges	Usually removes isolated dots
Text edges	Can blur edges	Usually preserves edges better
Filter type	Linear	Non-linear

6.1 Worked numerical example

Consider a 3 × 3 window with these values:

Input window (two corrupted values)

12	15	14
255	13	0
16	15	14

Sorted values: 0, 12, 13, 14, 14, 15, 15, 16, 255.
Median: the fifth value = 14.
Average: (12 + 15 + 14 + 255 + 13 + 0 + 16 + 15 + 14) / 9 = 354 / 9 ≈ 39.3 (about 39).
The median result, 14, is close to the normal neighbouring values; the average is pulled away by 0 and 255.

7. Answer to (e): Step by step with a full example

Step 1: Observe the scanned page. Locate the small black and white dots.

Step 2: Choose the filter. Select a 3 × 3 median filter for isolated salt-and-pepper noise.

Step 3: Place the window. Position the window around a noisy pixel.

Step 4: Sort the nine values. Arrange the values in ascending order.

Step 5: Find the fifth value. Choose the median; in the example it is 14.

Step 6: Correct the pixel. Replace the corrupted centre value with 14.

Step 7: Repeat and inspect. Process the remaining pixels and check that text edges remain clear.

Median-filter process



8. Short exam-ready answer

(a) Salt-and-pepper noise consists of random black (0) and white (255) pixels.
(b) It makes text difficult to read, damages letters and causes OCR errors.
(c) Use a 3 × 3 median filter.
(d) It ignores isolated extreme values and preserves edges better than averaging.
(e) Place a 3 × 3 window, sort its nine values, select the fifth value, replace the centre pixel, and repeat.

Conclusion: A median filter is suitable for removing isolated salt-and-pepper noise from scanned documents while retaining clearer text edges.

9. Extra viva questions

Question	Short answer
Is the median filter linear?	No. It is a non-linear filter.
What does 0 represent?	Black in a standard 8-bit grayscale image.
What does 255 represent?	White in a standard 8-bit grayscale image.
Why use an odd window size?	It provides one exact middle value.
Can a larger window be used?	Yes, for heavier noise, but thin lines and small details may be lost.

Question 4: Blurred Edges Scenario

Scenario: After noise removal, the image looks too smooth and the edges have become blurry.

1. The scenario and the questions

(a) Why did the edges become blurry?
(b) What problems come from over-smoothing?
(c) Which technique restores the edges?
(d) Why did you choose it?
(e) Write your answer step by step.

2. Basic ideas you must know first

Edge: An edge is a boundary where image brightness changes sharply, such as the border of a printed letter, building, road or object.

Smoothing: A smoothing filter reduces rapid brightness changes and small noise by averaging or suppressing high-frequency detail. Too much smoothing also weakens real edges.

Sharpening: An enhancement method that increases local contrast around boundaries so they appear clearer. It can improve apparent sharpness, but cannot perfectly restore details that were permanently lost.

3. Answer to (a): Why did the edges become blurry?

Edges become blurry when a smoothing or averaging filter combines pixels from both sides of a boundary. Dark pixels and bright pixels are mixed, creating intermediate gray values. This reduces the contrast at the boundary and spreads the transition over several pixels.

Illustration: sharp edge values

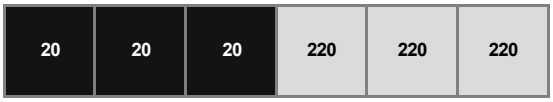
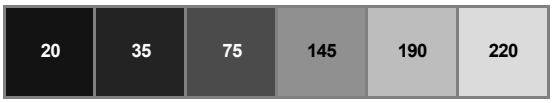


Illustration: smoothed edge values



In the first row, brightness changes suddenly from 20 to 220. In the second row, the gradual change shows a blurred edge.

4. Answer to (b): What problems come from over-smoothing?

• Small features, thin lines and fine textures may disappear.

• Boundaries between objects become unclear.

• Nearby objects may appear joined together.

• Edge detection and image segmentation may become less accurate.

• Satellite, medical or document images may lose details needed for interpretation.

• Repeated sharpening after strong smoothing can create halos or amplify remaining noise.

5. Answer to (c): Which technique restores the edges?

Use **edge sharpening**, particularly **unsharp masking**. It creates a slightly blurred version of the image, subtracts that blurred version from the original to obtain a detail mask, and adds a controlled amount of the mask back to the original.

5.1 Unsharp masking formula

Sharpened image = Original image + k × (Original image − Blurred image), where **k** controls sharpening strength. A moderate value is preferred; excessive sharpening can create halos and amplify noise.

6. Answer to (d): Why choose unsharp masking?

Unsharp masking is easy to understand and widely used. It enhances local contrast around edges, making letters and object boundaries appear clearer. The sharpening strength can be adjusted to balance edge improvement and noise

amplification. It is chosen because it targets detail differences rather than simply increasing brightness everywhere.

Technique	Benefit	Limitation
Unsharp masking	Simple, adjustable edge enhancement	Can create halos or amplify noise if too strong
Laplacian sharpening	Emphasises rapid intensity changes and edges	May strongly amplify noise
More smoothing	Can reduce remaining noise	Usually makes the edge blur worse

6.1 Worked example

Suppose a pixel has original intensity 150 and the corresponding blurred intensity is 130. The detail mask is 150 − 130 = 20. If k = 1, the sharpened value is 150 + 20 = 170. This increases contrast at that location. In a real image, values are clipped to the valid range (0–255) if necessary.

Quantity	Value
Original pixel	150
Blurred pixel	130
Detail mask	150 − 130 = 20
Sharpening strength k	1
Output	150 + (1 × 20) = 170

7. Answer to (e): Step by step

Step 1: Inspect the image. Identify the boundaries that have become soft after noise removal.

Step 2: Create a blurred copy. Apply a mild blur to the image to estimate its smooth background.

Step 3: Calculate the detail mask. Subtract the blurred copy from the original image.

Step 4: Control the strength. Multiply the detail mask by a suitable factor k.

Step 5: Add details back. Add the scaled mask to the original image to enhance local edge contrast.

Step 6: Check for artifacts. Look for halos, excessive contrast or amplified noise and reduce k if needed.

Step 7: Compare results. Compare the original, smoothed and sharpened versions to ensure the result is clearer.

Unsharp masking process



8. Short exam-ready answer

(a) Edges blur because smoothing mixes pixel values on both sides of a boundary.
(b) Over-smoothing removes fine details, weakens boundaries and reduces recognition accuracy.
(c) Use edge sharpening, such as unsharp masking.
(d) It increases local edge contrast and has adjustable strength.
(e) Blur a copy, subtract it from the original, scale the detail mask, add it back, and check for halos.

Conclusion: Controlled unsharp masking can improve apparent edge sharpness after smoothing, but it cannot recreate information that has been permanently lost.

9. Extra viva questions

Question	Short answer
What is an edge?	A location with a sharp change in image intensity.

Question	Short answer
Does sharpening recover all lost detail?	No. It improves apparent sharpness but cannot perfectly recreate removed information.
What happens if k is too high?	Halos and noise amplification may occur.
Why is unsharp masking named that way?	A blurred (unsharp) copy is used to create a detail mask.
Should sharpening be applied strongly?	No. Use controlled sharpening and compare the result.

Question 5: High-Frequency Noise Scenario

Scenario: A satellite image has fine grain-like noise that hides small features.

1. The scenario and the questions

(a) What is high-frequency noise?
(b) What problems does it cause?
(c) Which frequency-domain filter can be used?
(d) Why did you choose this filter?
(e) Write your answer step by step.

2. Basic ideas you must know first

Spatial domain: The image represented as pixel values arranged in rows and columns.

Frequency domain: A representation of how quickly image intensity changes across space. Low frequencies correspond broadly to gradual changes and large structures; high frequencies correspond to rapid changes such as fine edges, textures and also fine noise.

High-frequency noise: Unwanted rapid pixel-to-pixel variations that often appear as fine grain. Important warning: real edges and small features also contain high-frequency information, so removing all high frequencies would damage the image.

3. Answer to (a): What is high-frequency noise?

High-frequency noise is unwanted image variation that changes rapidly between neighbouring pixels. In a satellite image it can look like fine grain scattered over land, water, vegetation or buildings. It may arise from sensor electronics, acquisition conditions or transmission/processing effects.

4. Answer to (b): What problems does it cause?

• The image appears grainy and visual interpretation becomes harder.

• Small roads, boundaries, rooftops or vegetation patterns may be hidden by the grain.

• Texture measurements and feature extraction can become less reliable.

• Classification and segmentation algorithms may mistake noise for real surface detail.

• The noise may create false edges and reduce confidence in image analysis.

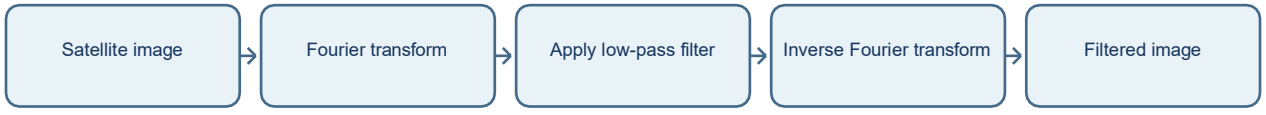
5. Answer to (c): Which frequency-domain filter can be used?

A **low-pass frequency-domain filter** can reduce high-frequency noise. One suitable choice is a **Butterworth low-pass filter**. A Gaussian low-pass filter is another option. The filter attenuates high frequencies while retaining low-frequency image content.

5.1 How the Butterworth low-pass filter works

The filter keeps frequencies near the centre of the shifted Fourier spectrum more strongly and gradually reduces frequencies beyond a chosen cutoff. Its order controls how quickly attenuation increases. A gradual roll-off helps avoid an abrupt cutoff, but the cutoff and order still need to be selected carefully.

Frequency-domain filtering process



6. Answer to (d): Why choose this filter?

Fine grain-like noise often contains high-frequency components, so a low-pass filter can reduce its strength. A Butterworth filter is useful because its cutoff frequency and order can be adjusted to balance smoothing and detail preservation. A moderate cutoff is important: a cutoff that is too low may remove real details such as narrow roads, building edges and small objects.

Filter	Main characteristic	Consideration
Butterworth low-pass	Adjustable cutoff and order; gradual frequency attenuation	A high order can produce a sharper transition; tune parameters carefully
Gaussian low-pass	Smooth attenuation with no abrupt frequency cutoff	May soften fine features
Ideal low-pass	Abruptly removes frequencies beyond cutoff	Can cause ringing artifacts around edges

6.1 Conceptual example

Assume a satellite image contains useful low-frequency landscape variation and unwanted fine grain concentrated in higher frequencies. A low-pass filter retains more of the broad landscape variation and reduces some of the grain. However, small roads and sharp boundaries may also be weakened because they contain high-frequency information.

Important: The Fourier spectrum is not simply “noise in the outer area and useful information in the centre.” The spectrum depends on the image. Frequency filtering reduces selected components; it cannot perfectly distinguish noise from all genuine details.

7. Answer to (e): Step by step

Step 1: Load the satellite image. Use the image that contains visible fine grain-like noise.

Step 2: Convert to frequency domain. Apply the two-dimensional Fourier transform to represent spatial patterns as frequency components.

Step 3: Shift the spectrum. Move the zero-frequency component to the centre for easier filter design and viewing.

Step 4: Select filter parameters. Choose a low-pass filter and a moderate cutoff frequency; select the Butterworth order if using that filter.

Step 5: Apply the filter. Multiply the Fourier spectrum by the low-pass filter mask to attenuate higher frequencies.

Step 6: Return to spatial domain. Apply the inverse Fourier transform and take the real image result.

Step 7: Evaluate the output. Compare the filtered image with the original and check noise reduction as well as preservation of small features.

7.1 Optional pseudocode

1. Read the satellite image.
2. Compute the 2-D Fourier transform.
3. Shift the spectrum so the low-frequency centre is visible.
4. Create a Butterworth low-pass mask using a chosen cutoff and order.
5. Multiply the spectrum by the mask.
6. Apply the inverse Fourier transform.
7. Display and compare original and filtered images.

8. Short exam-ready answer

(a) High-frequency noise is rapid pixel-to-pixel intensity variation, often seen as fine grain.
(b) It hides small features, creates false edges and affects classification or segmentation.
(c) Use a low-pass frequency-domain filter, such as a Butterworth low-pass filter.
(d) It attenuates high-frequency components and offers adjustable cutoff and roll-off.
(e) Apply Fourier transform, shift the spectrum, apply the low-pass mask, perform inverse transform and inspect the result.

Conclusion: A carefully tuned low-pass filter can reduce high-frequency grain in satellite imagery. The cutoff must be selected to avoid removing important small features.

9. Extra viva questions

Question	Short answer
What are low frequencies associated with?	Broad, gradual intensity changes and larger image structures.
What can high frequencies represent?	Fine detail, sharp edges, texture and high-frequency noise.

Question	Short answer
Does low-pass filtering remove only noise?	No. It may also remove real edges and small details.
Why choose a moderate cutoff?	To reduce grain while preserving as much useful detail as possible.
What is the inverse Fourier transform used for?	To convert the filtered frequency representation back to an image.

Final quick revision table

Question	Problem	Recommended method	Main reason
3	Black and white dots in a scanned document	3 × 3 median filter	Removes isolated extreme pixels and preserves text edges better than averaging
4	Edges become blurry after smoothing	Unsharp masking / controlled sharpening	Increases local contrast around boundaries
5	Fine grain-like noise in satellite image	Butterworth or Gaussian low-pass filter in frequency domain	Attenuates selected high-frequency components; cutoff must preserve useful details

Overall note: No filter is perfect for every image. Select the method based on the type of noise, apply it with suitable parameters, and compare the result against the original to check whether important details have been preserved.